Individual Reflection – NewsBot Intelligence System

ITAI 2373 – Natural Language Processing
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Project: Midterm NewsBot Intelligence System

1. Which NLP techniques did you find most challenging? Which were most useful?

The hardest part for me was getting the different NLP techniques to work together. I ran into some errors while working with the machine learning models, especially when the data was not in the format the model expected. I had to go back, check my code, and figure out what was causing the problem.

I found text classification and Named Entity Recognition the most useful. It was interesting to see how the system could recognize different news categories and identify names, companies, and locations from an article.

2. How did you handle combining multiple NLP tasks?

I worked on the project step by step instead of trying to do everything at once. I started with text preprocessing, then moved to feature extraction, sentiment analysis, text classification, and Named Entity Recognition.

Once I had the different sections working, I put them together into the NewsBot system. Some parts didn't work correctly the first time, so I had to review my code and run the tests again. Working this way made it easier for me to find mistakes.

3. What real-world problems could this system solve?

I think a system like NewsBot could be useful for news websites that deal with a lot of articles every day. Instead of organizing everything manually, they could use it to automatically group articles into different categories.

Businesses could also use it to follow news about companies or products. The sentiment analysis could help them understand whether the coverage is positive or negative, while Named Entity Recognition could identify important people and organizations mentioned in the articles.

4. What are the potential risks of automated news analysis?

One thing I noticed during testing is that the system does not always give the correct results. For example, when I tested an article about climate change, the system classified it as Business with low confidence. It also gave the article a positive sentiment even though the topic involved serious concerns.

That surprised me because the code was running, but the prediction was not really accurate. It made me realize that just because a model produces an answer doesn't mean the answer is correct. Bias in the training data can also affect predictions. I think these systems still need human review, especially when the information is important.

5. What NLP topics are you most excited to explore next?

I would like to learn more about topic modeling and language models like BERT. Topic modeling sounds interesting because it can help find common topics in a large collection of documents without having to read every single one.

I am also curious about how newer language models understand text and whether they can handle some of the problems I noticed while testing NewsBot.

6. How did you divide the work and ensure quality?

I worked on this project independently, so I had to handle the coding, testing, analysis, and documentation myself.

I followed the notebook sections and completed them one at a time. After running the code, I checked the outputs to make sure they made sense. I also tested the final NewsBot system using three different articles to see how it performed.

I learned that testing is important because even when the code runs without errors, the results can still be wrong.

7. How will you present this project to potential employers?

I would add this project to my GitHub portfolio so employers can see some of the things I have learned in Natural Language Processing.

I would explain how NewsBot can classify articles, analyze sentiment, and identify important information in text. I would also show the results from the different models, including the Naive Bayes model that reached 93.75% accuracy.

I think showing the actual code and results would be a good way to demonstrate what I have practiced in this class.

Final Reflection

This project was challenging for me, especially when I had to fix errors and get the different parts of NewsBot working together. There were times when the code worked, but the results were not what I expected.

One of the biggest things I learned is that getting an output from a machine learning model doesn't always mean the prediction is correct. I need to pay attention to the results and try to understand why the model makes certain predictions.

Overall, I feel more comfortable working with Python and NLP tools than I did before starting this project. I still have a lot to learn, but this project gave me more practice and helped me understand how the techniques we studied in class can be used together.